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Laporan Pengerjaan Backend Siger Pangan:

1. Perancangan Arsitektur Sistem & Microservices

- Merancang dan memecah backend menjadi beberapa microservice mandiri agar lebih scalable.

- Setiap service berjalan di dalam container Docker yang terisolasi. Service yang dibuat meliputi:

  a. Service Auth: Mengelola autentikasi (JWT & Google OAuth 2.0 Web/Mobile) dan Role-Based Access Control (RBAC).

  b. Service Catalog: Mengelola master data komoditas pangan dan wilayah/pasar.

  c. Service Web Scraper: Mengambil data harga pangan dari API Bank Indonesia secara otomatis menggunakan cron job dan BullMQ.

  d. Service NLP: Chatbot AI berbasis FastAPI dan Google Gemini untuk menjawab query natural language seputar harga.

- Menggunakan Nginx sebagai API Gateway untuk melakukan reverse proxy dan me-routing traffic ke masing-masing container microservice.

```
MINGW64:/c/Users/faqih0

faqih0@faqih MINGW64 ~
$ docker ps
CONTAINER ID   IMAGE                                   COMMAND                  CREATED             STAT
US                                  PORTS
                                   NAMES
8c3ca02abe8b   cloudflare/cloudflared:latest           "cloudflared --no-au…"   About a minute ago   Rest
arting (255) 28 seconds ago
                                   siger-cloudflare-tunnel
4d38d3320004   nginx:alpine                            "/docker-entrypoint.…"   About a minute ago   Up A
bout a minute                      0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
                                   siger-nginx-dev
4623675017a9   adminer:4-standalone                    "entrypoint.sh docke…"   About a minute ago   Up A
bout a minute                      0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
                                   siger-adminer-dev
9143f4b826f5   siger-pangan-dev-service-web-scraper    "docker-entrypoint.s…"   About a minute ago   Up A
bout a minute                      0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
                                   siger-web-scraper-dev
d048eb9e5ebb   siger-pangan-dev-service-auth           "docker-entrypoint.s…"   About a minute ago   Up A
bout a minute                      0.0.0.0:3001->3000/tcp, [::]:3001->3000/tcp
                                   siger-auth-dev
24d54af479b2   siger-pangan-dev-service-catalog        "docker-entrypoint.s…"   About a minute ago   Up A
bout a minute                      0.0.0.0:3002->3000/tcp, [::]:3002->3000/tcp
                                   siger-catalog-dev
152a66cd5b58   postgres:16-alpine                      "docker-entrypoint.s…"   About a minute ago   Up A
bout a minute (healthy)            0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
                                   siger-postgres-dev
7d8a87e5c8db   redis:7-alpine                          "docker-entrypoint.s…"   About a minute ago   Up A
bout a minute (healthy)            0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
                                   siger-redis-dev
5fc9defbeae0   siger-pangan-dev-service-nlp            "./entrypoint.sh"        About a minute ago   Up A
bout a minute                      0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp, 0.0.0.0:50051->50051/tcp, [::
]:50051->50051/tcp   siger-nlp-dev

faqih0@faqih MINGW64 ~
$
```

2. Setup Database & Caching

- Menggunakan PostgreSQL sebagai database utama, lengkap dengan skema dan seed data awal (15 kabupaten/kota Lampung, 16 komoditas).

- Mengimplementasikan Redis untuk caching pada setiap endpoint API dengan Time-To-Live (TTL) 1 jam agar response time lebih cepat dan mengurangi beban server.

Select: scraper_runs

Select data Show structure Alter table New item

Select | Search | Sort | Limit: 50 | Text length: 100 | Action: Select

SELECT * FROM "scraper_runs" LIMIT 50 (0.001 s) Edit

Modify	id	job_id	status	market_type_id	records_inserted	records_updated	records_skipped	date_range_start	date_range_end	error_message	duration_ms	started_at	completed_at
edit	54b82cbd-1ae9-43a6-9eab-b76e9328d954	6	success	NULL	168	0	0	2026-04-01	2026-04-11	NULL	2001	2026-04-11 12:49:25.762995	2026-04-11 12:49:27.761+00
edit	bce82314-3e89-4393-9199-c076a10c108a	1	success	NULL	32364	0	0	2024-04-18	2026-04-18	NULL	163652	2026-04-18 13:44:25.404197	2026-04-18 13:47:09.04+00
edit	4496a74e-6a2b-4a07-bcfe-ba1114c396d6	2	success	NULL	32364	0	0	2024-04-18	2026-04-18	NULL	175364	2026-04-18 13:47:09.051482	2026-04-18 13:50:04.414+00
edit	8df6841c-ec85-4137-b26e-f115df6fa720	daily-2026-04-23	success	NULL	372	0	0	2026-04-16	2026-04-23	NULL	10570	2026-04-23 23:00:00.031203	2026-04-23 23:00:10.581+00
edit	e5f4cb36-9856-403d-b1e9-fbe952cf0166	daily-2026-05-02	success	NULL	310	0	0	2026-04-25	2026-05-02	NULL	10294	2026-05-02 23:00:00.025609	2026-05-02 23:00:10.304+00

Whole result: 5 rows | Modify: Save | Selected (0): Edit Clone Delete | Export (5)

Import

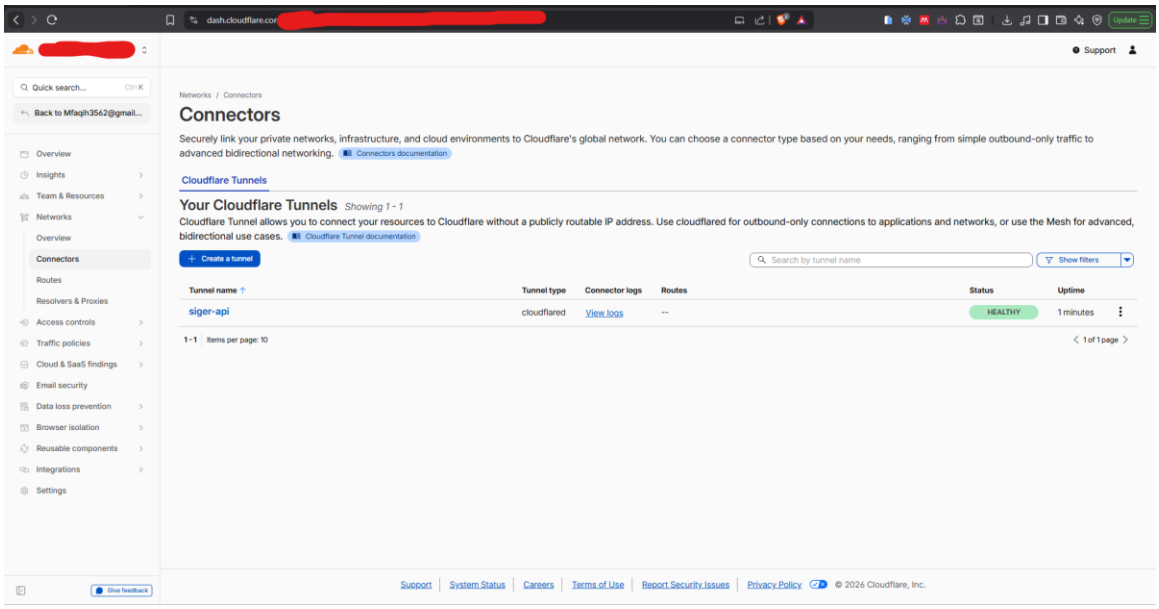
```
faqih0@faqih MINGW64 ~
$ docker exec siger-postgres-dev psql -U siger_user -d siger_pangan_dev -c "SELECT * FROM scraper_runs LIMIT 5;"

                  id                   |      job_id      | status  | market_type_id | records_inserted | records
_updated | records_skipped | date_range_start | date_range_end | error_message | duration_ms |         started_a
t          |         completed_at
---------------------------------------+------------------+---------+----------------+------------------+--------
---------+-----------------+------------------+----------------+---------------+-------------+------------------
----------+----------------------------
 54b82cbd-1ae9-43a6-9eab-b76e9328d954 | 6                | success |                |              168 |
       0 |               0 | 2026-04-01       | 2026-04-11     |               |        2001 | 2026-04-11 12:49:
25.762995 | 2026-04-11 12:49:27.761+00
 bce82314-3e89-4393-9199-c076a10c108a | 1                | success |                |            32364 |
       0 |               0 | 2024-04-18       | 2026-04-18     |               |      163652 | 2026-04-18 13:44:
25.404197 | 2026-04-18 13:47:09.04+00
 4496a74e-6a2b-4a07-bcfe-ba1114c396d6 | 2                | success |                |            32364 |
       0 |               0 | 2024-04-18       | 2026-04-18     |               |      175364 | 2026-04-18 13:47:
09.051482 | 2026-04-18 13:50:04.414+00
 8df6841c-ec85-4137-b26e-f115df6fa720 | daily-2026-04-23 | success |                |              372 |
       0 |               0 | 2026-04-16       | 2026-04-23     |               |       10570 | 2026-04-23 23:00:
00.031203 | 2026-04-23 23:00:10.581+00
 e5f4cb36-9856-403d-b1e9-fbe952cf0166 | daily-2026-05-02 | success |                |              310 |
       0 |               0 | 2026-04-25       | 2026-05-02     |               |       10294 | 2026-05-02 23:00:
00.025609 | 2026-05-02 23:00:10.304+00
(5 rows)


faqih0@faqih MINGW64 ~
$ |
```

3. Setup Domain & Cloudflare Tunnel

- Membeli dan melakukan setup custom domain pribadi untuk backend: `sigerpangan.my.id`.

- Menggunakan Cloudflare Tunnel (cloudflared) untuk mengekspos API lokal/server ke public internet dengan aman tanpa harus membuka port router secara manual (port-forwarding).

- Trafik dari domain dialihkan secara aman langsung ke Nginx API Gateway.

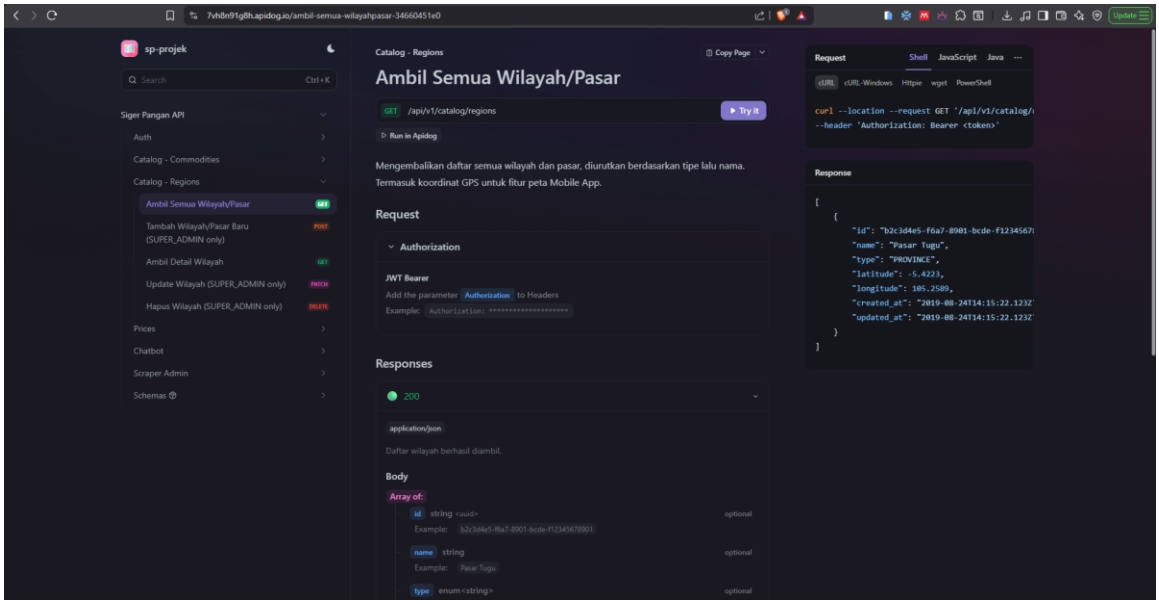


4. Dokumentasi API & Panduan Integrasi Frontend

- Menyusun dokumentasi spesifikasi OpenAPI/Swagger (siger-pangan-api.openapi.json) secara lengkap untuk seluruh endpoint.

- Membuat panduan integrasi sistem untuk tim Frontend Mobile (Flutter) dan Web CMS (Next.js) yang menjelaskan cara mengakses dan menampilkan data dari masing-masing API.

Link: 7vh8n91g8h.apidog.io



Tech Stack Keseluruhan:

NestJS, FastAPI, PostgreSQL, Redis, BullMQ, Docker, Nginx, Cloudflare Tunnel, gRPC, Google Auth Library, Google Generative AI (Gemini).